



A Multi-aspect Multi-granularity Pronunciation Assessment Method Based on Multi-feature Fusion and Transformer Encoder

Yawei Zhao^{1,2,3} , Aishan Wumaier^{1,2,3} , Xueliang Guo^{1,2,3} ,
and Yaxuan Lv^{1,2,3}

- ¹ College of Computer Science and Technology, Xinjiang University, Urumqi, China
Hasan1479@xju.edu.cn
- ² Xinjiang Multimodal Intelligent Processing and Information Security Engineering
Technology Research Center, Urumqi, China
- ³ Joint International Research Laboratory of Silk Road Multilingual Cognitive
Computing, Urumqi, China

Abstract. The Computer-Assisted Pronunciation Training (CAPT) system provides non-native (L2) speakers with an efficient learning path. As a core component of CAPT, Automatic Pronunciation Assessment (APA) quantifies learners' pronunciation abilities and offers precise feedback. However, current APA methods fail to model hierarchical dependencies across different granularities and lack robust speech representations. To address these issues, we propose the Cwacformer, a hierarchical multi-granularity pronunciation assessment model with a convolution-augmented Transformer encoder. Key innovations include: (1) word timing features for capturing rhythm and prosodic patterns, (2) a dual-branch architecture combining global Transformer and local CNN pathways with learnable fusion, (3) multi-scale convolution for enhanced word-level modeling, (4) adversarial training with Generative Adversarial Networks (GANs) to improve robustness, and (5) cross-attention mechanisms for explicit hierarchical relationship modeling. Extensive experiments on the Speechocean762 dataset show significant improvements across all granularities, with phoneme-level mean squared error (MSE) reduced to 0.073 and word-level stress recognition PCC reached 0.483, a 51% relative improvement over the baseline. Cwacformer achieves superior performance in most evaluation metrics, particularly excelling in stress detection and fluency assessment, while maintaining robustness across different speakers and acoustic conditions.

Keywords: Computer-Assisted Pronunciation Training ·
Pronunciation Assessment · Transformer

1 Introduction

Improving pronunciation is a key challenge for non-native (L2) learners, and Computer-Assisted Pronunciation Training (CAPT) systems aim to provide

effective solutions for this task. Both Mispronunciation Detection and Diagnosis (MDD) and Automatic Pronunciation Assessment (APA) are core components of CAPT systems. While MDD focuses on error identification, APA provides a comprehensive evaluation of pronunciation quality, covering aspects such as fluency, naturalness, intonation, stress, and rhythm. These systems are essential because they provide learners with personalized, real-time feedback, which is difficult to achieve in traditional classroom settings.

Current APA methods face three critical limitations: (1) insufficient modeling of hierarchical relationships between granularities, which leads to inconsistent cross-level predictions; (2) the lack of suprasegmental timing information, resulting in poor assessment of rhythm and stress; and (3) limited robustness to acoustic variations, causing performance degradation across different speakers and conditions. These limitations arise from fundamental architectural choices that treat granularities independently and ignore the temporal-prosodic structure of speech. Existing approaches either employ parallel architectures that overlook inter-granularity dependencies or use simplistic attention mechanisms that fail to capture complex contextual relationships between phonemes, words, and utterances.

To address these challenges, we propose Cwacformer, a multi-granularity pronunciation assessment model based on multi-feature fusion and Transformer encoders. Our key contributions are as follows: (1) novel word timing features that capture rhythm-aware suprasegmental information, (2) adversarial training mechanisms using GANs to enhance robustness, (3) cross-attention modules that explicitly model hierarchical relationships between granularities, and (4) comprehensive experimental validation demonstrating significant improvements across various evaluation aspects.

2 Related Work

2.1 Multi-granularity and Multi-aspect Assessment Evolution

Early research on Automatic Pronunciation Assessment (APA) focused on single-granularity evaluations [15, 18, 21, 22], offering a limited assessment scope. To provide a more comprehensive skill evaluation, multi-granularity methods were introduced to assess different linguistic levels (phoneme, word, and utterance). The GOPT model [10] was one of the pioneers of multi-dimensional prediction but used parallel architectures that overlooked hierarchical relationships. The Hipama model [8] introduced a hierarchical structure with multi-aspect attention but did not adequately model inter-granularity dependencies.

Traditional approaches focused on single aspects [1, 6, 19, 21], whereas real-world applications require the simultaneous assessment of accuracy, fluency, and intonation. Recent multi-aspect models have utilized multi-task learning frameworks: Ryu et al. [17] integrated APA and MDD with shared representations, and Chao et al. [2] developed HMamba, a model using hierarchical state space models with deXent loss.

2.2 Feature Enhancement Approaches

Deep learning approaches have advanced APA through self-supervised learning [7], ASR transfer learning [14], and end-to-end architectures [13]. Chao et al. [3] combined GOP features with self-supervised learning and suprasegmental information for multi-level assessment, demonstrating excellent performance but lacking explicit hierarchical modeling and robust representations.

Our work addresses these limitations by incorporating word timing features for suprasegmental modeling, adversarial training for robustness enhancement, and cross-attention mechanisms for explicit hierarchical relationship modeling.

3 Method

3.1 Overview

We introduce Cwacformer, a hierarchical model designed for multi-granularity pronunciation assessment. At the phoneme level, a lightweight Transformer encoder [20] is combined with an enhanced convolutional branch, which are learnably fused to capture both global context and local phonetic details. The model’s robustness is further improved through an adversarial perturbation module [11] during training. At the word level, multi-scale convolution and a tri-branch Bi-LSTM network are employed to capture local temporal cues and bidirectional context. At the utterance level, we replace word-feature averaging with concatenation and self-attention, and integrate word- and utterance-level information using cross-attention. This architecture effectively integrates multi-granularity modeling, local-global feature fusion, and robustness optimization, resulting in significant improvements in automatic pronunciation assessment (APA). The model structure is shown in Fig. 1.

3.2 Model Architecture

Model Inputs: For the model input, we follow the approach outlined in the baseline model [8] to ensure a fair comparison. The automatic speech recognition (ASR) acoustic module takes both the audio and its transcription as input and predicts frame-level phonetic posterior probabilities. These probabilities are then transformed into 84-dimensional GOP features, which consist of the following components:

$$x_{Gop} = [LPP(p_1), \dots, LPP(p_{42}), LPR(p_1|p_i), \dots, LRP(p_{42}|p_i)] \quad (1)$$

where LPP [12] and LPR [22] represent the log phone posterior and log posterior ratio for a phone p , respectively. A phone p refers to one of the pure phones in the acoustic model.

Traditional GOP-based features provide information at the phoneme segment level but fail to capture suprasegmental information adequately [5]. To address this limitation, we follow the input setup from the study [23], adding two additional features: duration x_{Dur} and energy x_{Eng} . While features like

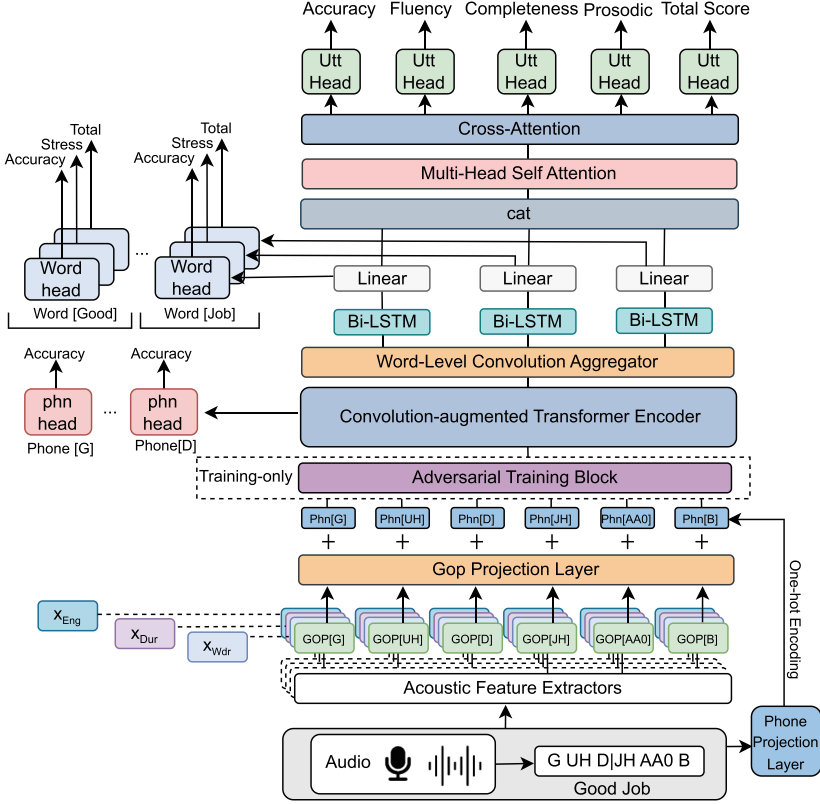


Fig. 1. Overall architecture of Cwacformer.

x_{Dur} and x_{Eng} offer a preliminary way to incorporate suprasegmental information, they often represent low-level acoustic properties rather than higher-level linguistic patterns. From a linguistic perspective, pronunciation quality emerges from the interaction between segmental accuracy (phoneme-level) and suprasegmental patterns (rhythm and stress). Therefore, to better bridge this gap, we propose novel word timing features x_{Wdr} , which encode duration patterns that are linguistically meaningful for stress and rhythm assessment. The specific steps are shown in Fig. 2.

$$\mu_i = \frac{1}{n_i} \sum_{j=s_i}^{e_i} d_j, \quad \sigma_i = \sqrt{\frac{1}{n_i} \sum_{j=s_i}^{e_i} (d_j - \mu_i)^2} \quad (2)$$

$$r_i = \max_{j \in [s_i, e_i]} d_j - \min_{j \in [s_i, e_i]} d_j, \quad \delta_i = \frac{1}{n_i - 1} \sum_{j=s_i}^{e_i-1} |d_{j+1} - d_j| \quad (3)$$

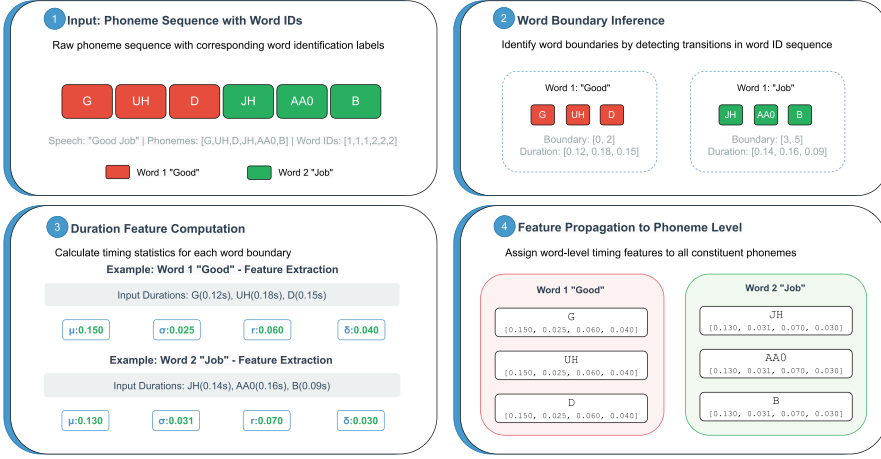


Fig. 2. Word Timing Features Extraction Process.

where d_j represents the duration of the j -th phoneme, s_i and e_i denote the start and end indices of the i -th word boundary, and $n_i = e_i - s_i + 1$ is the number of phonemes in the i -th word. The mean duration μ_i reflects the overall speech rate within the word, the standard deviation σ_i captures rhythm stability and duration variability, the duration range r_i measures the amplitude of duration variation, and the duration delta δ_i quantifies rhythmic smoothness by assessing fluctuations between adjacent phonemes. The timing feature vector for the i -th word is constructed as $\mathbf{f}_{timing}^{(i)} = [\mu_i, \sigma_i, r_i, \delta_i] \in R^4$. To maintain compatibility with phoneme-level processing, each word's timing features are propagated to all constituent phonemes: $\mathbf{x}_{Wdr}^{(j)} = \mathbf{f}_{timing}^{(i)}$ for all $j \in [s_i, e_i]$.

To maintain consistency with the baseline model [8], we apply one-hot encoding to the standardized phonemes and map them to a specified dimension to obtain the phoneme representation x_{Phn} . The input feature structure of the model is as follows:

$$x = [x_{Gop}; x_{Dur}; x_{Eng}; x_{Wdr}] \quad (4)$$

$$x_{input} = Linear(x) + x_{Phn} \quad (5)$$

where x_{Gop} represents the GOP features, x_{Dur} represents the duration features, x_{Eng} represents the energy features, x_{Wdr} represents the word timing features, x_{Phn} represents the one-hot encoding of the phonemes, and x_{input} is the final input to the model.

Adversarial Training Block: To enhance model robustness against input feature perturbations, we introduce an adversarial perturbation module [11] based on Generative Adversarial Networks (GANs). This module optimizes feature stability and generalization through Generator-Discriminator competition.

Generator employs a three-layer fully connected network to generate perturbation vectors from encoded features. Tanh activation constrains perturbations to $[-1, 1]$, which are added to original features creating “fake features”. The Generator aims to produce perturbations indistinguishable to the Discriminator, enhancing feature robustness.

Discriminator distinguishes between real and fake features, outputting scalar authenticity scores (1=real, 0=fake). It optimizes loss to effectively differentiate perturbations from real features, providing reverse optimization signals to the Generator.

During inference, the adversarial module is disabled for efficiency and consistency. The model relies solely on original features for scoring, eliminating training-time perturbation complexity while benefiting from enhanced robustness learned during adversarial training.

Convolution-Augmented Transformer Encoder: Accurately capturing both local details and global context is crucial for phoneme-level APA modeling. We employ a dual-branch architecture: a Transformer encoder captures long-range dependencies through self-attention for global semantic representations, while an enhanced gated convolutional block extracts local information and pronunciation anomalies, as illustrated in Fig. 3.

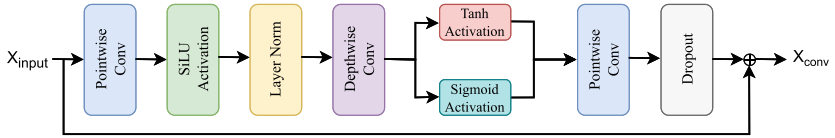


Fig. 3. Illustration of the Convolution Module. The enhanced convolutional branch employs pointwise convolution (kernel=1) to double feature dimensions, followed by SiLU activation and layer normalization. Depthwise convolution (kernel=3) enhances local context modeling. A gating mechanism multiplies the Tanh and Sigmoid outputs to dynamically adjust feature channels, highlighting key phoneme positions. After dimension restoration via pointwise convolution and dropout, a residual connection is applied.

Feature fusion uses a learnable parameter $\alpha \in [0, 1]$ (initialized at 0.5) to balance global and local features:

$$x_{self} = SelfAttn(x_{input}) + x_{input} \quad (6)$$

$$x_{conv} = Conv(x_{input}) + x_{input} \quad (7)$$

$$x_p = \alpha x_{self} + (1 - \alpha) x_{conv} \quad (8)$$

where x_{self} and x_{conv} represent Transformer and convolutional branch outputs, and x_p is the final encoder output with dynamically balanced global-local features.

Word-Level Convolution Aggregator: The baseline model [8] uses only multi-aspect attention at the word level, which considers features within a single granularity and inadequately models challenging aspects like stress. To address this limitation, we propose a multi-scale convolution structure to better capture local and neighboring contexts for word-level scoring. Multi-scale convolution is motivated by the need to capture pronunciation patterns at different temporal scales. Stress detection requires analyzing variations in phoneme duration, energy patterns, and pitch changes across different context windows. Small kernels (size 1) capture point-wise phoneme characteristics, medium kernels (size 3) model local phoneme transitions, and large kernels (size 5) capture broader prosodic patterns that are essential for stress identification. We apply three convolution kernels of different scales to the phoneme-level features:

$$x_{conv_i} = Conv_i(x_p), \quad i \in \{1, 3, 5\} \quad (9)$$

$$x_{Mconv} = concat(x_{conv_1}, x_{conv_3}, x_{conv_5}) \quad (10)$$

where x_p represents phoneme-level output features, x_{conv_i} are outputs from different kernel sizes, and x_{Mconv} is the concatenated multi-scale feature representation that enables comprehensive modeling of pronunciation patterns at multiple temporal resolutions.

Although multi-scale convolution captures local features from different receptive fields, temporal dependencies and long-range contextual relationships are equally crucial for APA. Word-level scoring depends not only on local phoneme combinations but also on surrounding context, prosodic patterns, and temporal information. Therefore, we introduce three parallel Bi-LSTM branches to process the multi-scale convolution outputs. Each Bi-LSTM processes the input sequence bidirectionally through input gates, forget gates, output gates, and memory cells, enabling a comprehensive analysis of temporal dependencies. The outputs are then transformed through linear layers for dimensional adaptation:

$$x_{wi} = Linear(BiLSTM_i(x_{Mconv})), \quad i = 1, 2, 3 \quad (11)$$

where x_{Mconv} represents the multi-scale convolution output, and x_{wi} represents the word-level features for accuracy, stress, and overall performance scoring, respectively.

Utterance-Level Modeling: The baseline model [8] averages word-level features (accuracy, stress, total score) before utterance evaluation, potentially obscuring feature differences and limiting multi-aspect modeling capabilities. We replace averaging with feature concatenation to preserve distinctiveness and enhance utterance-level expressiveness. To improve global context capture, we employ multi-head attention followed by cross-attention inspired by image classification [4]. The concatenated features serve as Query, while self-attended features act as Key and Value, enabling dynamic extraction of contextually relevant information across evaluation granularities. After residual connection, features

pass through five parallel linear modules for independent representations of accuracy, completeness, fluency, prosody, and total score, with separate regression heads completing utterance evaluation:

$$x_{cat} = \text{concat}(w_1, w_2, w_3) \quad (12)$$

$$x_{attn} = \text{Attention}(x_{cat}) \quad (13)$$

$$x_{cross} = \text{Attention}(x_{cat}, x_{attn}, x_{attn}) \quad (14)$$

$$x_u = \text{Linear}(x_{attn} + x_{cross}) \quad (15)$$

where w_1, w_2, w_3 represent word-level outputs, x_{cat} is the concatenated features, x_{attn} is self-attended output, and x_{cross} applies cross-attention with x_{cat} as Query and x_{attn} as Key/Value for utterance-level feature generation.

3.3 Loss

We employ mean squared error (MSE) as the primary loss function for pronunciation assessment tasks, following standard practice in the field. The total loss combines multi-granularity evaluation losses with adversarial training:

$$L_{total} = \sum_{m \in \{p, w, u\}} \lambda_m \frac{1}{N_m} \sum_{i=1}^{N_m} L_m^i + \lambda_{adv} L_{adv} \quad (16)$$

where L_m^i represents the MSE loss for the i -th aspect at granularity level m (phoneme, word, utterance), N_m is the number of aspects at each level, and λ_m are the loss weights. The adversarial loss encourages feature robustness through perturbation training:

$$L_{adv} = \text{BCE}(D(x + \tanh(\delta) \cdot \varepsilon), 1) \quad (17)$$

where $\delta = G(x)$ is the generator perturbation, $D(\cdot)$ is the discriminator, BCE denotes Binary Cross Entropy loss, and $\varepsilon = 0.03$ controls perturbation magnitude. We set $N_p = 1$, $N_w = 3$, $N_u = 5$, $\lambda_p = \lambda_w = \lambda_u = 1$, and $\lambda_{adv} = 0.1$.

4 Experiments

4.1 Experiments Setup

We evaluate our method on the Speechocean762 dataset [24], which contains 5,000 speech samples from 250 Mandarin-speaking L2 English learners (2,500 training/2,500 testing). Each sample is annotated by five professional reviewers across three granularities (utterance, word, phoneme) with scores normalized to $[0, 2]$. Model hyperparameters include: 1 Transformer Encoder layer, 1 Bi-LSTM layer, attention dimension 72, and embedding dimension 24. We employ a dual-optimizer strategy with Adam: Optimizer-G (lr=1e-3, 80 epochs with decay every 5 epochs after epoch 20) updates the backbone and generator, while

Optimizer-D ($\text{lr}=5\text{e-}3$) updates the discriminator. Training uses batch size 25 and GANs perturbation factor 0.03. Evaluation metrics are Pearson Correlation Coefficient (PCC) across granularities and Mean Squared Error (MSE) at phoneme level. Results report mean and standard deviation over five random seeds.

4.2 Result and Discussions

We compare our model with the following four existing evaluation models: (1) The GOPT model from [10], which uses a Transformer encoder in a parallel architecture; (2) The hierarchical architecture model HiPAMA from [8], which uses multi-aspect attention; (3) A parallel architecture model named Gradformer designed in [16], which employs a Transformer encoder-decoder architecture for granularity decoupling; (4) The Bfhaformer model from [9], which employs an LSTM-augmented BranchFormer encoder to encode GOP features and reference phoneme features. All of the above models employ a multi-task learning approach for multi-aspect and multi-granularity scoring. We performed 4 model trainings on the training set, and the mean and standard deviation of the test set are shown in Table 1.

Table 1. Comparison of Cwacformer with other models

Model	Phoneme Score		Word Score (PCC)			Utterance Score (PCC)				
	MSE	PCC	Accuracy	Stress	Total	Accuracy	Comp	Fluency	Prosodic	Total
GOPT	0.085 ± 0.001	0.612 ± 0.003	0.533 ± 0.004	0.291 ± 0.030	0.549 ± 0.002	0.714 ± 0.004	0.155 ± 0.039	0.753 ± 0.008	0.760 ± 0.006	0.742 ± 0.005
HiPAMA	0.084 ± 0.001	0.616 ± 0.004	0.575 ± 0.004	0.320 ± 0.021	0.591 ± 0.004	0.730 ± 0.002	0.276 ± 0.177	0.749 ± 0.001	0.751 ± 0.002	0.754 ± 0.002
Gradformer	0.079 ± 0.001	0.646 ± 0.004	0.598 ± 0.006	0.334 ± 0.013	0.614 ± 0.006	0.732 ± 0.005	0.318 ± 0.139	0.769 ± 0.006	0.767 ± 0.004	0.756 ± 0.003
Bfhaformer	0.080 ± 0.001	0.646 ± 0.003	0.621 ± 0.005	0.386 ± 0.024	0.635 ± 0.003	0.737 ± 0.004	0.448 ± 0.136	0.770 ± 0.004	0.770 ± 0.004	0.760 ± 0.003
Cwacformer	0.073 ± 0.002	0.687 ± 0.008	0.669 ± 0.015	0.483 ± 0.038	0.686 ± 0.016	0.773 ± 0.007	0.341 ± 0.151	0.796 ± 0.002	0.794 ± 0.002	0.793 ± 0.004

As demonstrated in Table 1, our model achieves the highest PCC scores in both phoneme and word-level evaluation tasks, with detailed analysis below:

Phoneme-level Analysis: Our model achieves an MSE of 0.073 and PCC of 0.687, significantly outperforming Bfhaformer and previous models. Like Gradformer and Bfhaformer, we employ parallel branches to capture local and global information. Unlike Gradformer, we improved the convolution block for local information extraction, reduced Transformer encoder layers, and used learnable parameters to weight global-local feature fusion. This enables simultaneous processing of global phoneme information and local temporal features.

Word-level Analysis: Our model substantially outperforms others, particularly in stress evaluation, clearly surpassing Bfhaformer (0.483 vs 0.386). Multi-scale convolution extracts local temporal features for recognizing energy and duration variations, while Bi-LSTM captures contextual information from preceding and succeeding words, enabling better understanding of stress locations. Through adversarial training with Generator and Discriminator, the model generates more robust perturbation features while maintaining stability and accuracy under noise and pronunciation variations.

Utterance-level Analysis: Our model outperforms previous models in all metrics except completeness. Cross-attention fuses word-level Bi-LSTM contextual information with global utterance-level self-attention features, providing comprehensive representations for utterance evaluation. For completeness, the model emphasizes stress, pitch variation, and rhythm features at phoneme and word levels. However, completeness depends more on fluency and pause positions. The model may overlook utterance coherence and appropriate pauses, leading to suboptimal completeness performance (0.341).

4.3 Ablation Studies

To explore the specific factors that influence Cwacformer performance, we performed five ablation experiments to verify the impact of each component on overall model performance. Each experiment involved removing components from the baseline model and incrementally adding our model components from the previous experiment. The ablation results are presented in Table 2.

Table 2. Ablation result

Model	Phoneme Score		Word Score (PCC)			Utterance Score (PCC)				
	MSE	PCC	Accuracy	Stress	Total	Accuracy	Comp	Fluency	Prosodic	Total
Baseline	0.084 ±0.001	0.616 ±0.004	0.575 ±0.004	0.320 ±0.021	0.591 ±0.004	0.730 ±0.002	0.276 ±0.177	0.749 ±0.001	0.751 ±0.002	0.754 ±0.002
+ Word Timing Features	0.078 ±0.002	0.653 ±0.008	0.614 ±0.008	0.344 ±0.015	0.630 ±0.008	0.763 ±0.004	0.068 ±0.147	0.790 ±0.004	0.786 ±0.004	0.788 ±0.006
+ Adversarial Training Block	0.078 ±0.001	0.650 ±0.005	0.614 ±0.005	0.324 ±0.022	0.629 ±0.005	0.759 ±0.003	0.110 ±0.175	0.786 ±0.005	0.781 ±0.003	0.784 ±0.003
+ Convolution-augmented Transformer Encoder	0.073 ±0.001	0.685 ±0.005	0.667 ±0.005	0.313 ±0.005	0.682 ±0.005	0.771 ±0.003	0.191 ±0.101	0.795 ±0.003	0.790 ±0.002	0.792 ±0.002
+ Word-level Convolution Aggregator	0.073 ±0.001	0.682 ±0.006	0.659 ±0.008	0.350 ±0.010	0.674 ±0.008	0.771 ±0.006	0.191 ±0.115	0.795 ±0.004	0.790 ±0.005	0.792 ±0.007
+ Bi-LSTM	0.074 ±0.001	0.680 ±0.005	0.662 ±0.011	0.482 ±0.036	0.677 ±0.011	0.778 ±0.006	0.143 ±0.103	0.795 ±0.004	0.792 ±0.005	0.797 ±0.005
Cwacformer	0.073 ±0.002	0.687 ±0.008	0.669 ±0.015	0.483 ±0.038	0.686 ±0.016	0.773 ±0.007	0.341 ±0.151	0.796 ±0.002	0.794 ±0.002	0.793 ±0.004

Word Timing Features provide substantial improvements across granularities, with word-level gains of 3.9%, 2.4%, and 3.9% in three dimensions.

These features capture rhythmic and prosodic relationships by understanding phoneme positions within words and utterances, addressing traditional phoneme-isolated evaluation limitations. However, utterance-level completeness decreased as it depends more on overall structure than word duration patterns.

Adversarial Training Block enables realistic perturbation feature generation through Generator-Discriminator feedback, though improvements were modest when added to baseline architecture, suggesting requirement for component synergy.

Convolution-augmented Transformer Encoder improves phoneme-level accuracy by combining global and local information extraction through parallel Transformer and enhanced convolution branches, providing richer representations for word-level processing.

Word-level Convolution Aggregator and **Bi-LSTM** achieve 16% improvement in stress recognition. Multi-scale convolution captures local temporal features for identifying stress positions (typically associated with energy and duration variations), while Bi-LSTM models bidirectional context for understanding stress placement and inter-word dependencies.

Cross-attention establishes effective information flow between utterance and word-level features, handling global information and contextual dependencies while maintaining accuracy in complex speech environments. This enhances utterance fluency and prosody capture, achieving hierarchical modeling through dynamic feature fusion across granularities.

Ablation Experiment for Word Timing Features and Adversarial Training Block: To evaluate the effectiveness of the Cwacformer model components, we conducted an ablation study to analyze the impact of removing Word Timing Features and Adversarial Training Block.

Table 3. Ablation study results of Word Timing Features and Adversarial Training Block

Model	Phoneme Score		Word Score (PCC)			Utterance Score (PCC)				
	MSE	PCC	Accuracy	Stress	Total	Accuracy	Comp	Fluency	Prosodic	Total
Cwacformer	0.073	0.687	0.669	0.483	0.686	0.773	0.341	0.796	0.794	0.793
	±0.002	±0.008	±0.015	±0.038	±0.016	±0.007	±0.151	±0.002	±0.002	±0.004
w/o Word Timing Features	0.078	0.660	0.613	0.497	0.630	0.739	0.219	0.798	0.795	0.766
	±0.001	±0.005	±0.005	±0.011	±0.004	±0.003	±0.133	±0.010	±0.009	±0.003
w/o Adversarial Training Block	0.075	0.671	0.638	0.458	0.656	0.763	0.238	0.794	0.791	0.785
	±0.001	±0.007	±0.014	±0.025	±0.014	±0.008	±0.066	±0.002	±0.001	±0.006

From Table 3, word timing features demonstrate clear benefits for most pronunciation assessment dimensions. While stress recognition shows a marginal variation when timing features are removed, the overall trend across all metrics confirms the importance of these features for comprehensive pronunciation

assessment. The removal of adversarial training consistently degrades performance across all evaluation aspects, confirming its crucial role in enhancing model robustness and generalization capability.

5 Conclusion

This paper introduces Cwacformer, a multi-granularity pronunciation assessment model designed to enhance accuracy and robustness for L2 learners through hierarchical structure integration and GANs-based adversarial training. The model leverages Bi-LSTM, multi-scale convolution, and cross-attention mechanisms to improve evaluation across stress, fluency, and accuracy dimensions. Experimental results on the Speechocean762 dataset demonstrate Cwacformer’s superior performance, particularly in word-level stress evaluation. Comparative experiments show consistent outperformance of existing baselines, and ablation studies confirm the effectiveness of each component. Despite these advances, the model faces limitations in utterance-level completeness evaluation, especially in capturing intonation patterns and pause positioning. Future work will focus on addressing these issues through model architecture optimization, potentially incorporating adaptive learning and transfer learning techniques to improve generalization across various linguistic contexts.

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